Research Communication

# Anti-inflammatory activity and molecular mechanism of delphinidin 3-sambubioside, a *Hibiscus* anthocyanin

**Takayuki Sogo**[1]
**Norihiko Terahara**[2]
**Ayami Hisanaga**[1]
**Takuma Kumamoto**[1]
**Takaaki Yamashiro**[1]
**Shusong Wu**[1]
**Kozue Sakao**[1,3]
**De-Xing Hou**[1,3]**

[1]*Course of Biological Science and Technology, United Graduate School of Agricultural Science, Kagoshima University, Kagoshima, Japan*

[2]*Department of Food Science and Technology, Faculty of Health and Nutrition, Minami-Kyushu University, Miyazaki, Japan*

[3]*Department of Biochemical Science and Technology, Faculty of Agriculture, Kagoshima University, Kagoshima, Japan*

## Abstract

Delphinidin 3-sambubioside (Dp3-Sam), a *Hibiscus* anthocyanin, was isolated from the dried calices of *Hibiscus sabdariffa* L, which has been used for folk beverages and herbal medicine although the molecular mechanisms are poorly defined. Based on the properties of Dp3-Sam and the information of inflammatory processes, we investigated the anti-inflammatory activity and molecular mechanisms in both cell and animal models in the present study. In the cell model, Dp3-Sam and Delphinidin (Dp) reduced the levels of inflammatory mediators including iNOS, NO, IL-6, MCP-1, and TNF-$\alpha$ induced by LPS. Cellular signaling analysis revealed that Dp3-Sam and Dp downregulated NF-$\kappa$B pathway and MEK1/2-ERK1/2 signaling. In animal model, Dp3-Sam and Dp reduced the production of IL-6, MCP-1 and TNF-$\alpha$ and attenuated mouse paw edema induced by LPS. Our *in vitro* and *in vivo* data demonstrated that *Hibiscus* Dp3-Sam possessed potential anti-inflammatory properties. © 2015 BioFactors, 41(1):58–65, 2015

***Keywords:*** *delphinidin 3-sambubioside; delphinidin; cytokines; mouse paw edema; anti-inflammation*

***Abbreviations:*** *Dp3-Sam, delphinidin 3-sambubioside; LPS, lipopolysaccharide; ERK, extracellular signal-regulated kinase; FBS, fetal bovine serum; iNOS, inducible nitric oxide synthase; i.p., intraperitoneally; JNK, c-Jun N-terminal kinase; MAPK, mitogen-activated protein kinase; MEK, MAPK/ERK kinase; NF-κB, nuclear factor kappa-light-chain-enhancer of activated B cells; NO, nitric oxide; SAPK, stress-activated protein kinase; s.c., subcutaneously; SEK, SAPK/ERK kinase.*



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**Address for correspondence: De-Xing Hou, Ph.D., Faculty of Agriculture, Kagoshima University, 1–21-24 Korimoto, Kagoshima 890-0065, Japan. Tel. & Fax: +81-99-285-8649;*
*E-mail: hou@chem.agri.kagoshima-u.ac.jp.*

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## 1. Introduction

Delphinidin 3-sambubioside (Dp3-Sam) is a *Hibiscus* anthocyanin isolated from the dried calices of *Hibiscus sabdariffa* L (Malvaceae). The dried calices of *Hibiscus sabdariffa* L are an ingredient of local beverages and a Chinese herbal medicine used to treat hypertension, pyrexia, and liver damage although the pharmaceutical components are poorly defined. Experimental data have showed that Dp3-Sam could *in vitro* inhibit proliferation of human leukemia cells (HL-60) by inducing apoptosis, which was characterized by cell morphology, DNA fragmentation, activation of caspase-3, -8, and -9, and inactivation of poly(ADP)ribose polymerase (PARP) [1]. On the other hand, we recently found that anthocyanin aglycons, especially delphinidin (Dp) and cyanidin (Cy) could *in vitro* suppress the expression of inflammatory mediators such as cyclooxygenase (COX-2) and prostaglandin (PG) $E_2$ [2]. Thus, the possibility that *Hibiscus* anthocyanins have anti-inflammatory activity needs to be considered.

During inflammatory disease, the primary cells of chronic inflammation are macrophages that produce excess amounts of mediators such as nitric oxide (NO) and proinflammatory cytokines such as TNF-α, IL-6, and MCP-1 [3,4]. In inflammatory site, NO is usually generated from L-arginine by inducible nitric oxide synthase (iNOS) [5]. NO overproduction leads to various harmful responses including apoptosis and necrosis [6], as well as organ failure in autoimmune diseases [7]. TNF-α is involved in many different cellular processes, including production of numerous cytokines and acute phase proteins, thus contributes to many pathophysiologic processes [8,9]. IL-6 is known as an important mediator of fever and of the acute phase response. IL-6 can be secreted by macrophages in response to specific microbial molecules to initiate innate immune system [10]. MCP-1, a proinflammatory chemokine, is a potent chemoattractant capable of promoting monocyte recruitment into an inflammatory or pathological site. The recruited cells can produce more proinflammatory mediators to potentiate inflammation [11]. Therefore, these proinflammatory cytokines play a pivotal role in consequences of inflammation [12].

Although the cellular signaling pathways regulating the inflammation are very complicated, mitogen-activated protein kinase (MAPK) and NF-κB signaling have been suggested at least to key pathways in the expression of inflammatory mediators including iNOS and cytokines [3,4]. MAPK signal pathway includes extracellular signal-regulated kinase (ERK), p38 MAPK (p38) and c-Jun $NH_2$-terminal kinase (JNK) [13]. These kinases can be activated after phosphorylation by bacterial LPS [14]. Activation of MAPK contributes to the production of inflammatory mediators such as iNOS and cytokines in activated macrophages [15,16]. NF-κB pathway includes IκB kinase (IKK), I kappa B (IκB) and p65/p50. Many stimuli including LPS, cytokines, and virus activate NF-κB via several signal transduction pathways, leading to IKK α and β activation that then phosphorylate IκB [17,18]. IKK α and β are activated by phosphorylation which is stimulated by phosphorylation of TAK-1 [19,20]. IκB degradation releases NF-κB from the complex, which enters the nucleus to transactivate a large variety of target genes such as iNOS and cytokines [21].

Based on the properties of Dp3-Sam and the information of inflammation processes, we investigated the anti-inflammatory activity and molecular mechanisms using both cell and animal models in the present study. First of all, we used mouse macrophage-like cells (RAW264.7), which can be stimulated with LPS to mimic a status of infection and inflammation, to investigate the inhibition and molecular mechanisms of Dp3-Sam and Dp on the productions of iNOS/NO, TNF-α, IL-6 and MCP-1. Finally, we confirmed the *in vivo* anti-inflammatory effects using a mouse paw edema model.

# 2. Materials and Methods

## 2.1. Reagents and Cell Culture

Dp3-Sam was prepared from dried calices of *Hibiscus sabdariffa* L. as described previously [1] and Dp purified by HPLC was obtained from Extrasynthese (Genay, France) (Fig. 1). LPS (*Escherichia coli* Serotype 055:B5) was from Sigma (St. Louis, MO). The antibodies against iNOS, MEK1, and α-tubulin were from Santa Cruz Biotechnology Inc. (Santa Cruz, CA). Other antibodies used were from Cell Signaling Technology (Beverly, MA). RAW264.7 cells were purchased from the American Type Culture Collection (Manassas, VA) and cultured in Dulbecco's modified Eagle's medium (DMEM) containing 10% FBS and 4 mM L-glutamine.

## 2.2. Measurement of NO Production

NO production was measured as concentrations of nitrogen oxides in culture medium with Griess method [22]. In brif, RAW264.7 macrophage cells ($3 \times 10^5$ cells/well) were plated in 48-well plates. After incubation for 24 h, cells were starved by being cultured in serum-free medium for another 2.5 h to eliminate FBS influence. The cells were then treated with or without Dp3-Sam or Dp for 30 min before exposure to 40 ng/mL LPS for 12 h. The accumulation of nitrites in culture medium was determined by measuring absorbance at 550 nm.

## 2.3. Measurement of Cytokine Production

Cytokine production was measured by Enzyme-Linked Immuno-Sorbent Assay (ELISA). In brief, RAW264.7 macrophage cells ($1.2 \times 10^5$ cells/well) were plated in 12-well plates. After incubation for 24 h, cells were starved by being cultured in serum-free medium for another 2.5 h to eliminate FBS influence. The cells were then treated with or without Dp3-Sam or Dp for 30 min before exposure to 40 ng/mL LPS for 12 h. The culture medium was used to assay the cytokine production with mouse ELISA Ready-SET-Go kit (eBioscience, San Diego, CA) according to manufacturer's instructions.

## 2.4. Western Blotting

Western blotting assay was performed as described previously [22]. Briefly, RAW264.7 cells ($1 \times 10^6$ cells/well) were cultured in 6 cm dishes for 24 h, cells were starved by being cultured in serum-free medium for another 2.5 h to eliminate influence of FBS. The cells were then treated with or without Dp3-Sam or Dp for 30 min before exposure to LPS for the indicated times. The cells were lysed in a lysate buffer, and boiled for 5 min. Approximate 40 μg of proteins were run on 10% SDS-PAGE and then transferred to PVDF membrane (GE Healthcare, UK). The blotted membrane was incubated with specific primary antibody for overnight at 4°C and further incubated for 1 h with HRP-conjugated secondary antibody. Bound antibodies were detected by ECL agent and further quantified by Lumi Vision Imager software (TAITEC Co., Japan).

## 2.5. *In Vivo* Paw Edema Model

The animal experiments were conducted in accordance with the guidelines of the Animal Care and Use Committee of Kagoshima University. Male ICR mice (4 weeks old) from Japan SLC Inc. were group-house under controlled light (12 h light/day) and temperature (25 °C). All the animals had free access to water and feed in a home cage. The mice were randomly divided into four groups: control, LPS, LPS plus

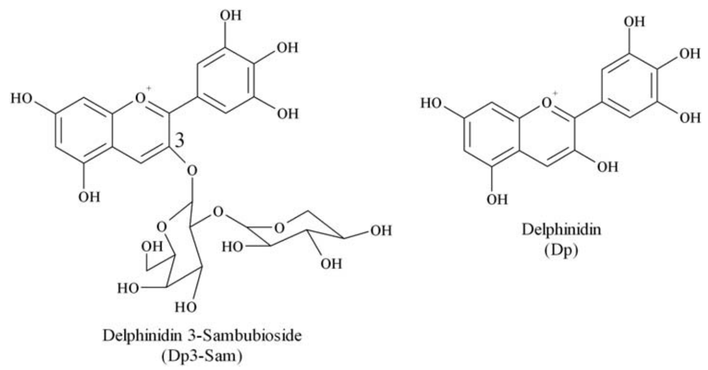


FIG 1 Chemical structures of delphinidin 3-sambubioside (Dp3-Sam) and delphinidin (Dp).

Dp3-Sam and LPS plus Dp. Dp3-Sam and Dp were dissolved in PBS containing 2% DMSO and administered to the mice by *i.p.* injection at 15 μMl/kg body weight for 4 days before LPS treatment. LPS were injected *i.p.* and *s.c.* to paw with 100 μL of 0.3 mg/mL (1 mg/kg body weight) and 10 μL of 3 mg/mL (1 mg/kg body weight) respectively. Paw thickness was measured using caliper (model 19975, Shinwa Rules Co. Ltd) before and every hour after LPS treatment until 6 h. After 6 h, mice were sacrificed and blood serum was collected for cytokines assay.

### 2.6. Statistical Analysis

All data were statistically analyzed by one way ANOVA and student *t*-test software in Excel form. Differences were considered significant for $P < 0.05$.

## 3. Results

### 3.1. The Inhibitory Effects of Dp3-Sam and Dp on the Production of Inflammatory Factors

To investigate the effects of Dp3-Sam and Dp on the production of inflammatory factors such as NO/iNOS, TNF-α, IL-6, and MCP-1. RAW264.7 cells were treated with 50 to 200 μM of Dp3-Sam or 100 μM of Dp for 30 min, respectively, before expose to 40 ng/mL LPS for 12 h. As shown in Fig. 2A, LPS-induced NO production was significantly suppressed by Dp3-Sam and Dp. And Dp3-Sam and Dp also inhibited LPS-induced iNOS expression in a dose-dependent manner (Fig. 2B). Next, we used ELISA assay to determine the production of TNF-α, IL-6, and MCP-1 in the same treatment. The data revealed that productions of these cytokines induced by LPS were significantly suppressed by Dp3-Sam and Dp (Fig. 2C), and Dp showed stronger inhibitory activity than Dp3-Sam.

### 3.2. Modulation of Dp3-Sam and Dp on MAPK Signaling

MAPK signaling is one of the important cell signaling pathways that regulates cytokines and proinflammatory mediators such as IL-6, TNF-α, MCP-1, and iNOS during inflammatory response. Thus, we investigated the effects of Dp3-Sam and Dp on the LPS-induced phosphorylation of MAPKs in RAW264.7 cells. The cells were treated with 50 to 200 μM Dp3-Sam or 100 μM Dp for 30 min, respectively, before exposure to 40 ng/mL LPS for 30 min. As shown in Fig. 3, treatment with 40 ng/mL LPS for 30 min caused phosphorylation of JNK1/2, ERK1/2, p38 kinase. Pretreatment with 100 μM Dp for 30 min suppressed markedly their phosphorylation while pretreatment with 50 to 200 μM Dp3-Sam only markedly suppressed ERK1/2 phosphorylation with little effect on the phosphorylation of JNK1/2 and p38. To confirm this, we further examined the effects on the phosphorylation of MEK1/2, an upstream kinase of ERK1/2. The same inhibitory effect was also observed in the phosphorylation of MEK1/2. These data indicate that the downregulation of MEK/ERK is at least involved in the inhibition of LPS-induced inflammation by Dp-3-Sam.

### 3.3. Modulation of Dp3-Sam and Dp on NF-κB Signaling

Accumulated data indicate that NF-κB is one of the principal factors for the production of iNOS and proinflammatory cytokines [19,23]. Thus, we examined whether Dp3-Sam and Dp inhibits the degradation of IκB. As shown in Fig. 4, LPS markedly stimulated the degradation of IκB when RAW 264.7 cells were treated with 1 μg/mL LPS for 10 min. Pretreated with Dp3-Sam (100–200 μM) and Dp (100 μM) for 30 min effectively suppressed the degradation of IκB. Dp3-Sam or Dp alone showed no influence on these processes (data not shown). Simultaneously, the phosphorylation of p65, a major partner of NF-κB, was observed in LPS treatment alone, and the inhibition of p65 phosphorylation was observed in the treatment with Dp3-Sam (100–200 μM) and Dp (100 μM) (Fig. 4). Moreover, phosphorylation of IκBs is regulated by IκB kinases, IKKα and IKKβ [24]. Thus, we next investigated the effect of Dp3-Sam and Dp on phosphorylation of IKKα/β in the same treatment. As shown in Fig. 4, LPS induced the phosphorylation of IKKα/β (lane 2) significantly, and Dp3-Sam (100–200 μM) and Dp (100 μM) inhibited the phosphorylation of LPS-induced IKKα/β without any effect on control protein. The data indicated that the downregulation of IKK-mediated NF-κB signaling pathways is involved in the inhibition of LPS-induced inflammation by Dp3-Sam or Dp.

### 3.4. Inhibition of LPS-Induced Paw Edema in Mice

To further confirm the anti-inflammatory effects of Dp3-Sam and Dp *in vivo,* we used the model of mouse paw edema induced by LPS. The mice were divided into four groups: control, LPS, LPS plus Dp3-Sam, and LPS plus Dp. Each group had three mice, respectively. Dp3-Sam or Dp was injected *i.p.* (15 μM/kg body weight) for 4 days before LPS treatment. LPS was injected *i.p.* and *s.c.* (1 mg/kg body weight), and the paw thickness was measured using digital caliper before and every hour after LPS treatment until 6 h (Fig. 5A). The results showed that LPS treatment increased the paw swelling to 0.44 mm at 2 h, and 0.36 mm after 6 h. Pretreatment with Dp3-Sam for 4 days reduced the LPS-induced paw thickness to 0.15 mm at 2 h, and 0.05 mm after 6 h. On the other hand, pretreatment with Dp for 4 days reduced the thickness of

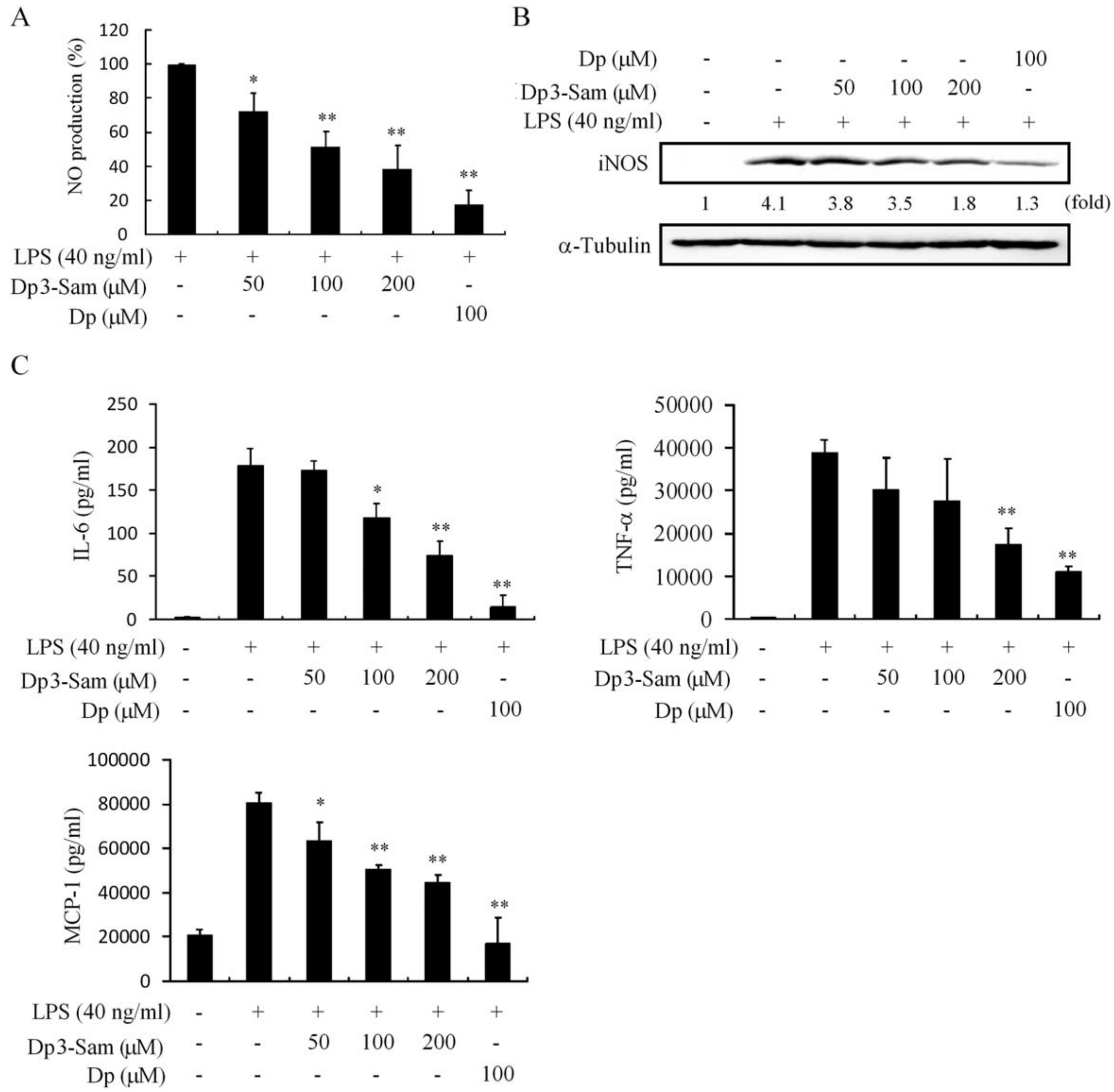


**FIG 2** *Influence of Dp3-Sam and Dp on the production of NO (A), iNOS (B), IL-6, TNF-α and MCP-1 (C). To detect NO and iNOS, RAW264.7 cells were precultured for 24 h, and starved in serum-free medium for 2.5 h. The cells were then treated with the indicated concentrations of Dp3-Sam or Dp for 30 min, and then exposed to 40 ng/mL LPS for 12 h. The indicated protein and α-tubulin were detected by Western blotting with their antibodies, respectively. The induction fold was calculated as the intensity of the treatment relative to that of control by densitometry. The blots shown are the examples of three separate experiments. To measure the level of IL-6, TNF-α, and MCP-1 (C), the cells (1.2 × 10$^5$ cells) were precultured for 24 h, and starved in serum-free medium for 2.5 h. The cells were then treated with the indicated concentration of Dp3-Sam or Dp for 30 min, and then exposed to 40 ng/mL LPS for 12 h. The amount of cytokines in medium was measured as described in Sections 2. Each value represents the mean ± S.D. of triplicate cultures. Different letter means *P < 0.05, **P < 0.01.*

LPS-induced paw thickness to 0.15 mm at 1 h, and the paw thickness was almost disappeared within 6 h. Dp3-Sam and Dp decreased the edema by 89.3% and 96.3%, respectively, compared with LPS alone after 6 h. As a control, treatment with PBS concluding 2% DMSO did not show any effect on paw edema (Fig. 5B). Simultaneously, we checked the levels of three serum cytokines, which were induced by LPS in RAW264.7 cells (Fig. 2), by ELISA. As shown in Fig. 5C, pretreatment with Dp3-Sam and Dp decreased the levels of LPS-induced serum IL-6, MCP-1 and TNF-α. These data confirmed *in vivo* the anti-inflammatory effect of Dp3-Sam and Dp that attenuated the production of LPS-induced inflammatory mediators to inhibit mouse paw edema.

# 4. Discussion

In the present study, we investigated the anti-inflammatory properties of a *Hibiscus* anthocyanin Dp3-Sam, comparing with its aglycon Dp, in both cell and animal models. In cell

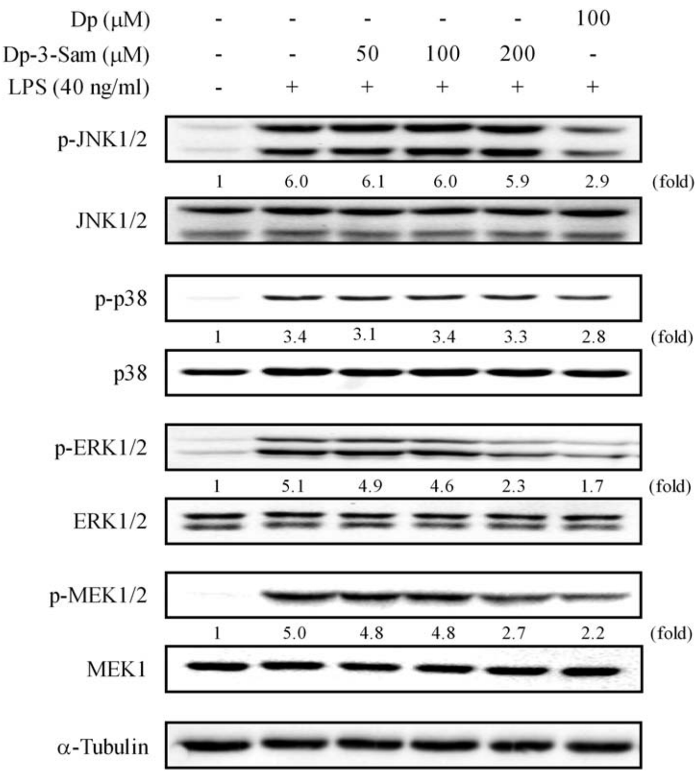


**FIG 3** *Dp3-Sam and Dp suppress the phosphorylation of ERK1/2 and MEK1/2. RAW264.7 cells were precultured for 24 h, and starved in serum-free medium for 2.5 h. After treatment with the indicated concentration of Dp3-Sam or Dp for 30 min, the cells were further exposed to 40 ng/mL LPS for 30 min. The total and phosphorylated protein kinases were detected with their antibodies, respectively. The induction fold of the phosphorylated kinase was calculated as the intensity of the treatment relative to that of control normalized to total protein or α-tubulin by densitometry. The blots shown are the examples of three separate experiments.*

model, mouse macrophage-like cell line (RAW264.7), which can be stimulated with LPS to mimic a status of infection and inflammation, was used to investigate the inhibitory effects and molecular mechanisms of Dp3-Sam and Dp on the productions of iNOS/NO, TNF-α, IL-6, and MCP-1. Dp3-Sam showed weaker inhibition on the production of these inflammatory factors than its aglycon Dp. However, in the animal model, almost equal inhibitory effect was observed in the inhibition of LPS-induced paw edema when Dp3-Sam or Dp was administrated with 15 μM/kg, *i.p.*

Our previous studies have showed that anthocyanidins, especially Dp and Cy, could suppress the expression of inflammatory mediators such as cyclooxygenase (COX-2) and prostaglandin (PG) E2 by attenuating cellular signaling including MAPK and NF-κB pathways [2]. However, most animal and human studies have found that anthocyanins present in natural food source are absorbed mainly in their intact glycosidic form, not aglyconic form [25–27]. For example, in rats, after a single oral administration of *Vaccinium myrtillus* anthocyanins, intact anthocyanins, but no anthocyanidin was detected in the plasma [28,29]. In the present study, we administered Dp to the mice by *i.p.* injection to avoid the absorption barrier by the gastrointestinal tract. Our data revealed that Dp could inhibit the production of inflammatory factors through the downregulation of cellular signaling including MAPK and NF-κB pathways. Several *in vitro* experiments showed that Dp is most strongest bioactive compound in the antioxidant [30,31], anti-cell transformation [30], and anti-inflammation activity [2] among six kinds of anthocyanidins. Moreover, MEK1 has been suggested as a direct target for Dp to suppress cell transformation [32]. The question is whether Dp3-Sam and Dp interfere with LPS recognition/binding of TLR4 to block LPS-induced inflammation signaling. For this, we have designed an experiment to investigate the binding of Dp3-Sam and Dp to TLR4 competitively with FITC-conjugated LPS by flow cytometric assay. Our primary data showed that Dp3-Sam and Dp did not reduce the density of FITC-conjugated LPS, suggesting that Dp3-Sam and Dp had no interfere with LPS recognition/binding of TLR4 (Hou et al., unpublished data).

Another interesting point is that Dp3-Sam could reveal the equal inhibitory effect as Dp on LPS-induced paw edema and inflammatory factors including TNF-α, IL-6 and MCP-1 although Dp3-Sam showed weaker inhibitory effects on these inflammatory factors than its aglycon Dp *in vitro*. The reason is unknown now. Several studies have suggested that the anthocyanins can be rapidly absorbed from the stomach after ingestion by a process that may involve bilitranslocase, and they enter the systemic circulation after passing through the

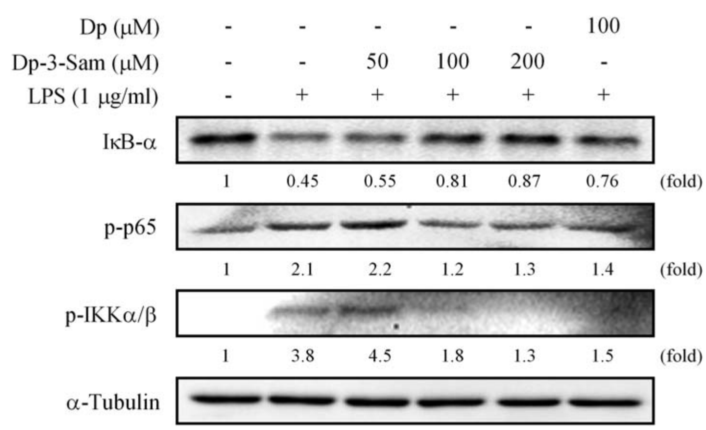


**FIG 4** *Dp3-Sam and Dp downregulate NF-κB signaling pathway. RAW264.7 cells were pretreated with the indicated concentration of Dp3-Sam or Dp for 30 min, and then exposed to 1 μg/mL LPS for 10 min. IκB-α, p-p65, p-IKKα/β, and α-tubulin were detected with their antibodies, respectively. The induction fold of the phosphorylated kinase was calculated as the intensity of the treatment relative to that of control normalized to α-tubulin by densitometry. The blots shown are the examples of three separate experiments.*

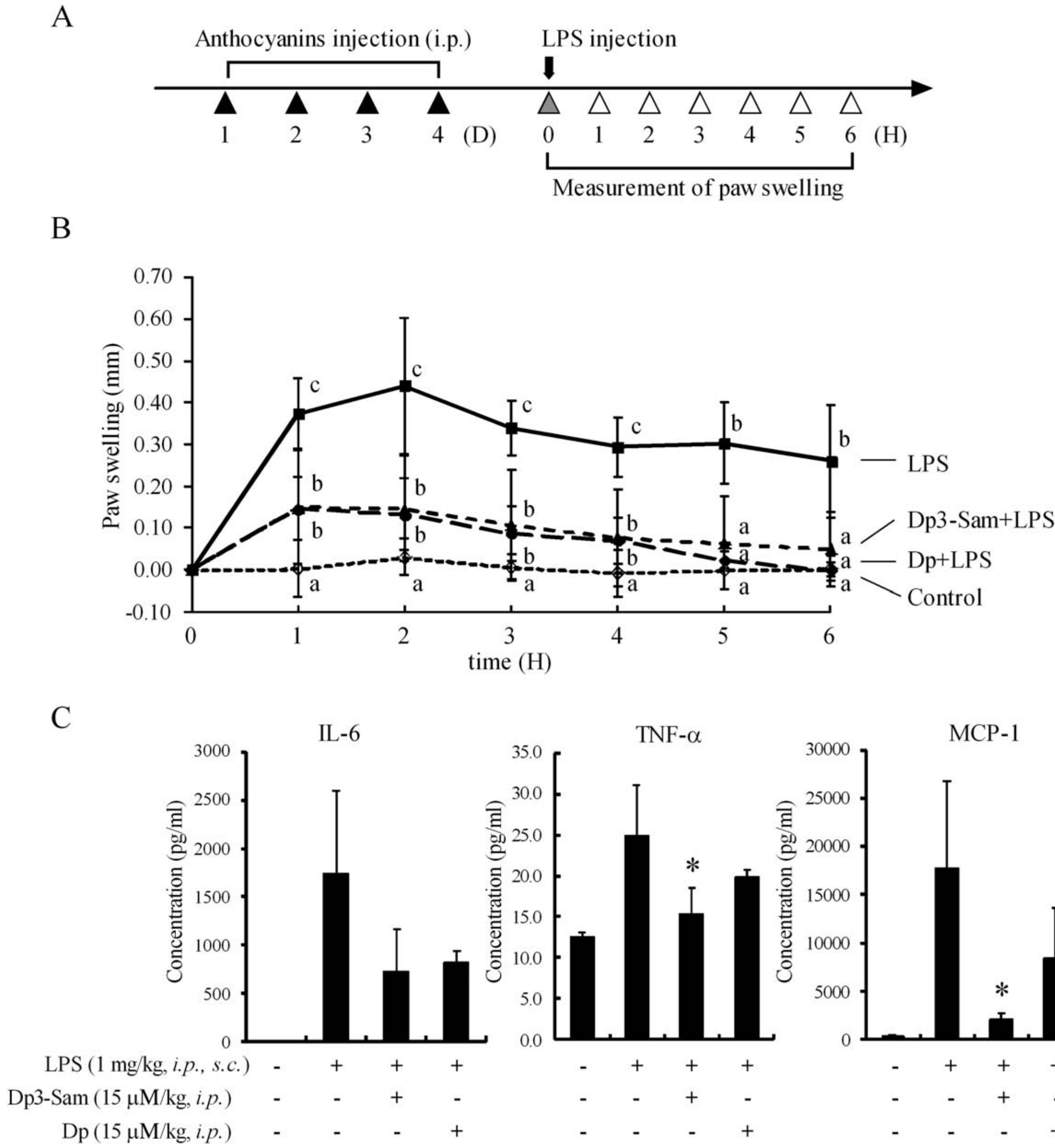


**FIG 5** *Dp3-Sam and Dp suppress paw edema in mice. The mice were divided into three groups: control, LPS, LPS + Dp3-Sam, and LPS + Dp. Each group had three mice, respectively. Dp3-Sam and Dp was injected i.p. (15 μM/kg body weight) for 4 days, and LPS was then injected i.p. and s.c. at paw (1 mg/kg body weight). The mouse paw thickness was measured using digital caliper before and every hour after LPS treatment until 6 h (A). The change of paw edema thickness was shown in (B). Means with differently lettered superscripts differ significantly at the probability of $p<0.05$. The change in level of serum IL-6, TNF-α, and MCP-1 is shown in (C). The blood serums were obtained from the mice that were treated with or without LPS for 6 h by collection of heart blood. The serum without any dilution was used for the measurement of IL-6, MCP-1, and TNF-α with mouse ELISA Ready-SET-Go kit (eBioscience) according to manufacturer's manual. Each value represents the mean ± S.D. of four mice. Different letter means *$P < 0.05$.*

liver [33]. Further absorption of anthocyanins that are not absorbed from the stomach move into the small intestine appears to take place in the jejunum. The transport mechanism involved has not been identified, but if similar to flavonols, may involve hydrolysis of the glycosides by various hydrolases and absorption of the phenolic aglycon [34]. In the present study, the administration of Dp3-Sam by *i.p.* injection is similar to the process in which anthocyanin glycosides is directly absorbed from the stomach, and the administration of Dp by *i.p.* injection appears similar to the process that anthocyanin is taken in the jejunum after the hydrolysis by various hydrolases and absorption of the phenolic aglycon. Moreover, the Phase II enzymatic hemi-synthesis of anthocyanin-*O*-methylated metabolites was observed by incubation with rat liver cytosolic proteins. Delphinidin-3-glucoside (Dp3G) was methylated to 4'-methylated Dp3G (Me-Dp3G) by phase II catechol-*O*-methyltransfarase (COMT), and Me-Dp3G has strong radical scavenging and anti-proliferative activity [33]. Moreover, some works confirmed that cyanidin-3-glucoside could be degraded to protocatechuic acid [35,36]. *In vitro* degradation product of Dp, gallic acid, could inhibit tumor cell growth [37]. Thus, it is important to clarify whether Dp3-Sam and Dp also produce these products after degradation to exert the biological activity in further work.

On the other hand, *i.p.* injection is not available for food-based strategies in most cases, and the results from *i.p.* injection may be different with that from oral administration. Thus, we like to purify sufficient amount of Dp3-Sam from *Hibiscus* and confirm the results by oral administration in future study. Although there are no direct report about the bioavailability and absorption of Dp and Dp3-Sam in animals and/or humans, some studies demonstrated that absorption and excretion vary between the different monoglucosides (galactoside, glucoside, and arabinosides) and aglycons (delphinidin, cyanidin, petunidin, peonidin, malvidin). For example, relatively more delphinidin glycoside is absorbed than malvidin glycoside, and relatively more galactoside is absorbed than arabinoside in rat when urinary excretion served as an indication of absorption [26,29]. A recent study on determination of the pharmacokinetic of 15 bilberry anthocyanins also demonstrated that among the aglycons, plasma concentrations were higher for the delphinidin and cyanidin than for the malvidin. Among glycosides, plasma concentrations were highest for galactosides and lowest for arabinosides [38]. These data suggested that delphinidin have higher bioavailability than other aglycons, and Dp3-Sam may have lower bioavailability which is worthy to clarify in future study.

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